AGN X-ray Heating: Derivation of the Emissivity-to-Heating-Rate Pipeline

Fatma Yousef

August 26, 2026

This note derives, from the AGN spectral model down to the T_K evolution equation, every equation implemented in `ComputeTs.c/XRayHeatingFunctions.c` for the AGN X-ray channel, in parallel with the pre-existing stellar (GAL) pathway. Every symbol is cross-referenced to its code variable so this can be checked line-by-line against the source. The specific question under discussion why the soft and hard bands need distinct integration limits in the *luminosity-conversion* step but not, for the same reason, in `integrate_over_nu` is addressed explicitly in Section 4.1. A second thread whether the AGN grids can be folded into the existing SFR grids to cut RAM is addressed in Section 6.

Contents

1 The AGN spectral model

Each band (soft, hard) is modelled as an independent power law in specific luminosity, pivoted at a threshold frequency $\nu_{\rm th}$ (`NuXrayThreshold`):

$$L_b(\nu) = L_{0,b}\left(\frac{\nu}{\nu_{\rm th}}\right)^{-\alpha_b}, \qquad b \in \{\text{soft}, \text{hard}\}, \tag{1}$$

where α_b is `SpecIndexXrayAGNSoft` or `SpecIndexXrayAGNHard`, and $L_{0,b}$ is the (as yet unknown) amplitude at the pivot. The two bands are defined over disjoint frequency ranges, split at the break frequency $\nu_{\rm break}$ (`NuXraySoftCut`):

$$\text{soft:} \quad \nu \in [\nu_{\rm th},\, \nu_{\rm break}], \tag{2}$$

$$\text{hard:} \quad \nu \in [\nu_{\rm break},\, \nu_{\rm max}], \qquad \nu_{\rm max} = \texttt{NuXrayMax}. \tag{3}$$

Each band's amplitude $L_{0,b}$ is fixed independently, by requiring that integrating Eq. (1) over that band's own frequency range reproduces a physically known, calibrated band luminosity $L_{{\rm band},b}$ (the accretion-derived L_X from the black-hole growth history, computed in `blackhole_feedback.c quasar_lx`, obscuration-corrected):

$$\int_{\nu_{{\rm lo},b}}^{\nu_{{\rm hi},b}} L_b(\nu)\, d\nu = L_{{\rm band},b}. \tag{4}$$

2 Step 1: the luminosity conversion factor

Define the *luminosity conversion factor* as the ratio of the pivot amplitude to the calibrated band luminosity,

$$C_b \equiv \frac{L_{0,b}}{L_{{\rm band},b}}, \tag{5}$$

which is exactly `Luminosity_converstion_factor_AGN_soft / _hard` in the code (up to the unit conversions in Section 2.3). Solving Eq. (4) for $L_{0,b}$ gives C_b.

2.1 Case $\alpha_b \neq 1$

$$L_{{\rm band},b} = L_{0,b}\,\nu_{\rm th}^{\alpha_b} \int_{\nu_{{\rm lo},b}}^{\nu_{{\rm hi},b}} \nu^{-\alpha_b}\, d\nu = L_{0,b}\,\nu_{\rm th}^{\alpha_b}\,\frac{\nu_{{\rm hi},b}^{1-\alpha_b} - \nu_{{\rm lo},b}^{1-\alpha_b}}{1-\alpha_b}, \tag{6}$$

$$\Rightarrow \quad C_b = \frac{(1-\alpha_b)\,\nu_{\rm th}^{-\alpha_b}}{\nu_{{\rm hi},b}^{1-\alpha_b} - \nu_{{\rm lo},b}^{1-\alpha_b}}. \tag{7}$$

Code check (soft band, `ComputeTs.c:1027-1037`):

```
Luminosity_converstion_factor_AGN_soft =
    pow(nu_hi, 1 - alpha) - pow(nu_lo, 1 - alpha);      // (nu_hi^(1-a) - nu_lo^(1-a))
Luminosity_converstion_factor_AGN_soft = 1.0 / ...;     // 1 / (...)
Luminosity_converstion_factor_AGN_soft *=
    pow(nu_th, -alpha) * (1 - alpha);                   // x nu_th^-a (1-a)
```

which is exactly Eq. (7), per band, with $(\nu_{{\rm lo},b}, \nu_{{\rm hi},b}) = (\nu_{\rm th}, \nu_{\rm break})$ for soft and $(\nu_{\rm break}, \nu_{\rm max})$ for hard.

2.2 Case $\alpha_b = 1$ (the singular case)

When $\alpha_b \to 1$, $1 - \alpha_b \to 0$ and Eq. (7) is 0/0: the power-law primitive $\nu^{1-\alpha}/(1-\alpha)$ degenerates, because $\int \nu^{-1} d\nu = \ln \nu$ instead. Re-deriving directly for $\alpha_b = 1$:

$$L_{\mathrm{band},b} = L_{0,b}\, \nu_{\mathrm{th}} \int_{\nu_{\mathrm{lo},b}}^{\nu_{\mathrm{hi},b}} \nu^{-1}\, d\nu = L_{0,b}\, \nu_{\mathrm{th}}\, \ln\left(\frac{\nu_{\mathrm{hi},b}}{\nu_{\mathrm{lo},b}}\right), \tag{8}$$

$$\Rightarrow \quad C_b = \frac{1}{\nu_{\mathrm{th}}\, \ln(\nu_{\mathrm{hi},b}/\nu_{\mathrm{lo},b})}. \tag{9}$$

Code check (`ComputeTs.c:1021-1026`):

```
if (fabs(SpecIndexXrayAGNSoft - 1.0) < REL_TOL) {
    Luminosity_converstion_factor_AGN_soft =
        (NuXrayThreshold*NU_over_EV) * log(NuXraySoftCut / NuXrayThreshold);
    Luminosity_converstion_factor_AGN_soft = 1.0 / ...;
}
```

which is exactly Eq. (9). This confirms the branch is not an arbitrary numerical safeguard: $\alpha_b = 1$ is a genuine mathematical singularity of the power-law primitive (a log-divergence of the naive formula), not just a coding edge case, so it needs its own closed-form solution rather than a tolerance-based patch of Eq. (7). The same structure (generic $\alpha \neq 1$ branch + $\alpha = 1$ log branch) already exists for GAL and, under `USE_MINI_HALOS`, for PopIII the AGN soft/hard code simply repeats it per band.

2.3 Unit conversion

C_b as derived above is dimensionless (it is a pure ratio of frequency-integrated shape to the same shape's own normalization). The code additionally divides by `PLANCK` (Section `ComputeTs.c:1042,1066`): this converts a specific-luminosity amplitude into a specific *photon*-luminosity amplitude ($L_\nu/h\nu_{\mathrm{th}}$-type normalization), consistent with everything downstream (`integrate_over_nu`'s integrand, Section 4) being expressed per unit photon energy. Unlike the GAL/PopIII conversion factors, no `SEC_PER_YEAR` factor is applied: GAL/III start from a star-formation-rate calibration ($M_\odot\,\mathrm{yr}^{-1}$) and need that factor to convert to a per-second rate; the AGN band luminosities are already luminosities ($\mathrm{erg\,s}^{-1}$), so no such conversion is needed.

3 Step 2: the redshift-dependent prefactor

The pivot amplitude, once fixed by C_b, still needs to be turned into the proper prefactor multiplying the frequency-shape integral at each emission redshift z' (`zp`). This is `const_zp_prefactor_AGN_soft/_hard`:

$$\mathcal{A}_b(z') = \frac{C_b}{\nu_{\mathrm{th}}}\, c\, (1+z')^{\alpha_b+3}. \tag{10}$$

Code check (`ComputeTs.c:1112-1130`):

```
const_zp_prefactor_AGN_soft = Luminosity_converstion_factor_AGN_soft
    / (NuXrayThreshold * NU_over_EV) * SPEED_OF_LIGHT
    * pow(1 + zp, SpecIndexXrayAGNSoft + 3);
```

which is Eq. (10) directly. This functional form $(1+z')^{\alpha+3}$ is inherited unmodified from the pre-existing GAL prefactor (itself the standard 21cmFAST/TS.C form of Pritchard & Furlanetto 2007 / Mesinger, Furlanetto & Cen 2011): the +3 comes from proper number-density dilution $\propto (1+z')^3$, and the α_b comes from redshifting a $\nu^{-\alpha_b}$ spectrum (each photon's frequency, and hence its place on the power law, redshifts along with z'). The AGN channel does not re-derive this it reuses the identical cosmological scaling, per band, with that band's own α_b and C_b.

Both prefactors are checked for finiteness immediately (`ComputeTs.c:1116,1127`) and the run aborts rather than silently propagating a bad value this guards against a zero-width band ($\nu_{\mathrm{hi},b} = \nu_{\mathrm{lo},b}$, e.g. `NuXraySoftCut==NuXrayThreshold`), which would make C_b divide by zero.

4 Step 3: the frequency-shape integral (`integrate_over_nu`)

This is a *different* integral, doing a *different* job. Where C_b (Section 2) fixes one number the overall amplitude of the band, so that the model matches a known calibration luminosity `integrate_over_nu` computes, at every radial shell R (i.e. every look-back shell z'', `zpp`) and every ionization state x_e, the deposition-efficiency-weighted photon-number shape kernel:

$$\mathcal{K}_b(z', z'', x_e) = \int_{\nu_{\mathrm{lo}}(z',z'')}^{\nu_{\mathrm{hi}}(z',z'')} f_{\mathrm{dep}}(\nu, x_e) \left(\frac{\nu}{\nu_{\mathrm{th}}}\right)^{-\alpha_b-1} d\nu, \tag{11}$$

where f_{dep} is the frequency- and ionization-dependent deposition/ionization/Lyα-production efficiency (`interp_fheat`, etc. inside `integrand_in_nu_heat_integral` and its ionization/Lyα siblings) and the extra power of ν^{-1} relative to Eq. (1) converts specific *luminosity* into specific *photon-number* flux ($L_\nu/h\nu$). This kernel is a pure numerical integral over whatever limits are handed to it it performs no calibration and needs no conversion factor, because nothing about it is required to reproduce an independently-known total.

4.1 Why its soft/hard limits differ from C_b's

Both places use different limits for soft vs. hard, but the two splits answer two different physical questions, which is the direct answer to "why are they needed here but not for `integrate_over_nu`":

- **C_b's limits** ($[\nu_{\mathrm{th}}, \nu_{\mathrm{break}}]$ vs. $[\nu_{\mathrm{break}}, \nu_{\mathrm{max}}]$, fixed, rest-frame) exist because *calibration is band-specific by definition*: C_b is only meaningful once "the band" is fixed, since it is derived by demanding $\int_{\mathrm{band}} L_b(\nu)\, d\nu$ equals that band's own known luminosity (Eq. (4)). Soft and hard are calibrated against physically distinct observables (soft-band and hard-band L_X, with distinct obscuration corrections and, potentially, distinct α_b), so they necessarily use distinct, fixed limits. This step happens once per redshift step, with no z'' dependence at all.

- **`integrate_over_nu`'s limits** (`lower/upper_int_limit_AGN_soft/hard`, recomputed every (z', z'') pair) exist for an unrelated reason: they track *which part of the rest-frame spectrum can physically still be soft or hard by the time it arrives at z'*. A photon emitted at z'' in the rest-frame hard band redshifts by $(1+z')/(1+z'')$ on its way to z', and can redshift down into what is, at z', the soft band hence the

  ```
  * (1. + zp) / (1. + zpp)
  ```

  factor on every AGN soft/hard limit (`ComputeTs.c:796-819`), absent from GAL/III's limits because GAL/III use one single band with no internal soft/hard split to redshift across. On top of that redshift scaling, the lower edge of each band is further clamped to $\max(\cdot,\ \nu_{\tau=1}(z', z''))$, the frequency at which the IGM becomes optically thick along that shell (`nu_tau_one_zpp`) a purely radiative-transfer constraint with no analogue in the calibration step at all.

In short: C_b's limits define *what the band is*, once, in the rest frame, for calibration purposes. `integrate_over_nu`'s limits define *what fraction of each band is currently visible*, recomputed every shell, for radiative-transfer purposes. They are both "different limits for soft vs. hard" but answer unrelated questions one is about the definition of a band, the other about a shifting horizon within it.

5 Step 4: assembling the heating-rate ODE term

Per shell z''_{ct} (in `evolveInt`, `XRayHeatingFunctions.c:1409-1423`):

$$I_b(z'', x_e) = X_b(z'')\,(1+z'')^{-\alpha_b}\,\mathcal{K}_b(z', z'', x_e), \tag{12}$$

$$\left.\frac{dx_{\mathrm{heat}}}{dt}\right|_b = \mathcal{A}_b(z') \sum_{z''} \frac{dt}{dz''}\,\Delta z''\ I_b(z'', x_e), \tag{13}$$

where $X_b(z'')$ is `XAGN_soft/_hard`[z"], the AGN band emissivity for that shell (an `SMOOTHED_AGN_(soft/hard)` grid value, itself built from `gal->BHXrayEmissivity` in `blackhole_feedback.c`). The $(1+z'')^{-\alpha_b}$ factor here is the emission-side counterpart of the $(1+z')^{\alpha_b+3}$ in $\mathcal{A}_b(z')$ (Eq. (10)) together they implement the same $(1+z'/1+z'')^{\alpha_b}$ spectral redshifting used to shift the `integrate_over_nu` limits in Section 4.1, applied this time to the amplitude rather than to the integration bounds. dt/dz'' is `dtdz(zpp)`, $\Delta z''$ is the shell width (`dzpp`), and the sum runs over `TsNumFilterSteps` radial/temporal shells.

Soft and hard are summed only after each has its own $\mathcal{A}_b$ applied (`XRayHeatingFunctions.c:1457-1469`):

$$\left.\frac{dx_{\mathrm{heat}}}{dt}\right|_{\mathrm{AGN}} = \left.\frac{dx_{\mathrm{heat}}}{dt}\right|_{\mathrm{soft}} + \left.\frac{dx_{\mathrm{heat}}}{dt}\right|_{\mathrm{hard}}, \tag{14}$$

and this combined AGN heating rate enters dT_K/dz' (`deriv[3] / dansdz[3]`) exactly where the stellar term does, alongside adiabatic cooling and Compton heating (unchanged from the pre-existing GAL/III pathway).

6 Can the AGN grids be folded into the existing SFR grids?

A separate review thread (`reionization.c`, on the `grids->BHXrayEmissivity` allocation block) raises the RAM cost of carrying `BHXrayEmissivity/_soft` as their own grids, history ring buffers, and FFTW plans, and asks whether the AGN contribution can instead be added into the existing SFR grid machinery by "manipulating the constants." Eq. (13) answers this precisely, by showing which quantities are genuinely band-specific and which are not.

6.1 The raw emissivities cannot be summed before filtering

Write the GAL contribution in the same form as Eq. (13):

$$\left.\frac{dx_{\mathrm{heat}}}{dt}\right|_{\mathrm{GAL}} = \mathcal{A}_{\mathrm{GAL}}(z') \sum_{z''} \frac{dt}{dz''}\,\Delta z''\ X_{\mathrm{GAL}}(z'')\,(1+z'')^{-\alpha_{\mathrm{GAL}}}\,\mathcal{K}_{\mathrm{GAL}}(z', z'', x_e). \tag{15}$$

Comparing this term-by-term with Eq. (13) for band b, every factor multiplying $X(z'')$ the redshifting exponent $(1+z'')^{-\alpha_b}$, the shape kernel $\mathcal{K}_b$ (itself α_b-dependent through the $(\nu/\nu_{\rm th})^{-\alpha_b-1}$ integrand in Eq. (11)), and the prefactor $\mathcal{A}_b(z')$ depends on α_b, which is a *different, independently-set parameter* for each of GAL/AGN-soft/AGN-hard (`SpecIndexXrayGal` vs. `SpecIndexXrayAGNSoft` vs. `SpecIndexXrayAGNHard`). Summing the raw grids first,

$$X_{\rm merged}(z'') \equiv X_{\rm GAL}(z'') + X_{\rm AGN}(z''), \tag{16}$$

and then applying *one* $(1+z'')^{-\alpha}\mathcal{K}_\alpha$ to $X_{\rm merged}$ is only correct if $\alpha_{\rm GAL} = \alpha_{\rm AGN}$ otherwise it silently applies the wrong spectral shape to whichever component's true α_b differs from the one used, exactly the failure mode raised in the reply on that thread. $X_{\rm GAL}$ and $X_{\rm AGN}$ are not even the same physical quantity before this step: $X_{\rm GAL}$ (`SMOOTHED_SFR_GAL`) carries units of SFR ($M_\odot\,{\rm yr}^{-1}$, converted to a luminosity only via $C_{\rm GAL}$, which bakes in $\alpha_{\rm GAL}$), while $X_{\rm AGN}$ (`SMOOTHED_AGN(_soft)`) is already a luminosity from `gal->BHXrayEmissivity(_soft)`. So merging *before* the calibration/shape stage i.e. merging the grids themselves is not just inconvenient but dimensionally and physically wrong whenever $\alpha_{\rm GAL} \neq \alpha_{\rm AGN,soft/hard}$, which is the general case here (they are independent fit parameters, not tied together). This confirms the concern raised in reply to that thread.

6.2 What genuinely is band-agnostic: the smoothing machinery

The real RAM cost the thread is pointing at, however, is not $X_b(z'')$ itself but the *scratch space used to compute it*: for each band, `ComputeTs.c` allocates its own pair of FFTW complex buffers (`*_unfiltered`/`*_filtered`, each $\mathcal{O}$(`ReionGridDim`^{3})) and forward/reverse plans, used only to smooth that band's raw per-cell emissivity over radius R (`filter()`, `ComputeTs.c:456-532`). Inspecting that pipeline shows every band GAL, III, AGN-hard, AGN-soft, AGN-LW goes through an *identical*, purely geometric sequence (forward FFT → normalize → copy → `filter()` → inverse FFT) that has no α_b dependence at all: `filter()` only ever sees a real-space grid of shape `ReionGridDim`^{3} and a radius R, never a spectral index. All α_b-dependence lives entirely upstream (Sections 2–4, in $C_b/\mathcal{A}_b/\ \mathcal{K}_b$) and downstream (Section 5, where each band's own smoothed history is weighted by its own α_b). Currently every band's forward FFT happens before any band's filtered result is extracted (all $N_{\rm bands}$ unfiltered/filtered pairs alive at once, so that one shared triple loop over cells can populate all the `SMOOTHED_*` arrays together, `ComputeTs.c:536-613/638-687`).

Because the filtering step itself is band-agnostic, the *scratch* buffers and plans not the `SMOOTHED_*` outputs could legitimately be shared: process one band fully (forward FFT → filter → inverse FFT → extract into that band's own `SMOOTHED_*` array), then reuse the same pair of complex buffers and the same FFTW plan for the next band, instead of holding up to five separate pairs simultaneously. This would cut the scratch-buffer cost from $\sim N_{\rm bands}\times$ down to $1\times$ $\mathcal{O}$(`ReionGridDim`^{3}), at the price of splitting the current shared extraction loop into one pass per band (more loop overhead, no change in numerical result). It does *not* touch the `SMOOTHED_*` history arrays themselves (Section 6.1 established those must stay physically distinct and they, not the scratch buffers, are the dominant memory cost, scaling as `ReionGridDim`^{3} $\times$ `TsNumFilterSteps` rather than just `ReionGridDim`^{3}), nor the per-galaxy history ring buffers (`bh_xray_histories(_soft)`), which record a physically distinct per-snapshot quantity from `sfr_histories` and cannot be merged for the same reason as Section 6.1.

7 Symbol / code cross-reference

Symbol	Meaning	Code name
$\nu_{\rm th}$	pivot frequency	`NuXrayThreshold`
$\nu_{\rm break}$	soft/hard split	`NuXraySoftCut`
$\nu_{\rm max}$	hard band ceiling	`NuXrayMax`
α_b	band spectral index	`SpecIndexXrayAGNSoft / Hard`
$L_{{\rm band},b}$	calibrated band luminosity	`quasar_lx` (obscuration-corrected)
C_b	luminosity conversion factor	`Luminosity_converstion_factor_AGN_soft/hard`
$\mathcal{A}_b(z')$	redshift prefactor	`const_zp_prefactor_AGN_soft/hard`
$\mathcal{K}_b(z', z'', x_e)$	frequency shape kernel	`freq_int_heat_tbl_AGN_soft/hard`
$X_b(z'')$	per-shell AGN emissivity	`XAGN_soft/hard[R_ct]`
$\nu_{\tau=1}(z', z'')$	opacity horizon	`nu_tau_one_zpp`

8 Open items for the meeting

- Confirm the sign/exponent convention in Eq. (1) ($L_\nu \propto \nu^{-\alpha}$) matches the intended definition of `SpecIndexXrayAGNSoft/Hard` as used elsewhere in the paper/thesis.
- Confirm whether $L_{{\rm band},b}$ (the calibration target) should itself already be obscuration-corrected at the point it enters Eq. (4), or whether obscuration is meant to apply only downstream (via `NHfrac/NHTrans`) this derivation assumes the latter (obscuration is applied separately, not baked into C_b).
- Verify the $\alpha_b = 1$ branch's tolerance (`REL_TOL` $= 10^{-5}$, `meraxes.h:65`) is tight enough not to introduce a discontinuity in C_b near $\alpha_b = 1$ for realistic parameter sweeps.
- Section 6: agree whether the scratch-buffer-reuse refactor (band-agnostic FFT buffers/plans, one band processed fully before reusing the pair for the next) is worth implementing given the RAM pressure at high `ReionGridDim`, and whether the resulting per-band extraction loop's extra overhead is acceptable.